Optimal Fiscal Policy in a Business Cycle Model

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November 9, 2026

Roadmap

1. Introduction
2. Decentralized Economy
3. Planning Problem
4. An Analytical Result
5. Quantitative Analysis

Introduction

Why study optimal fiscal policy over the business cycle?

- A central macroeconomic question is how distortionary fiscal policy should respond to aggregate fluctuations.
- Different policy frameworks deliver different prescriptions:
  - **Countercyclical fiscal policy:** tax cuts during recessions.
  - **Tax smoothing:** adjust taxes when shocks change the government's intertemporal budget constraint.
- CCK (1994) revisit this question in a stochastic neoclassical growth model with distortionary labor- and capital-income taxes.

The paper's question

How should the government optimally distribute tax distortions across time and states of nature when households accumulate capital and the government can issue state-contingent debt?

Motivation follows the paper's Introduction: CCK (1994, pp. 618–620).

Where does the paper fit in the literature?

Reference	Role in the discussion
Barro (1979)	Tax-smoothing benchmark: taxes respond to shocks that alter the government's present-value budget constraint.
Lucas and Stokey (1983)	State-contingent government debt can smooth tax distortions across states in an economy without capital.
Judd (1985); Chamley (1986)	Deterministic long-run benchmark: zero capital-income taxation in steady state.
Zhu (1992)	In a stochastic growth setting, there is no general theoretical presumption that labor taxes are constant or ex-ante capital taxes are zero.
Judd (1989)	A benchmark for the dynamics of ex-post capital taxes; CCK emphasize that their behavior also depends on the choice of policy within an equivalence class.

CCK's approach

Use a quantitative neoclassical business-cycle model to assess the importance of transition dynamics, tax smoothing, and state-contingent fiscal instruments.

Literature characterization is based on CCK (1994, Introduction).

The analytical architecture

1. **Decentralized economy:** households choose consumption, labor, capital, and government bonds; the government chooses distortionary taxes and debt returns.
2. **Primal Ramsey problem:** replace the full policy implementation problem with an implementability constraint and the aggregate resource constraint.
3. **Policy recovery:** recover labor taxes, ex-ante capital taxes, and taxes on private assets from the Ramsey allocation.
4. **Quantitative implementation:** solve stationary allocation rules conditional on the implementability multiplier and then determine that multiplier from initial conditions.

A key distinction

The Ramsey allocation can be uniquely characterized even when the ex-post capital tax and government-bond return are not separately determined.

Decentralized Economy

Environment and preferences

- Time is discrete: $t = 0, 1, \ldots$. Each period one of finitely many events s_t is realized.
- The history through period t is $s^t = (s_0, \ldots, s_t)$, with probability $\mu(s^t)$. The initial realization s_0 is given.
- Capital is predetermined at the beginning of period t, while labor and the shock are period-t objects:

$$k_t = k(s^{t-1}), \qquad n_t = n(s^t).$$

- A representative household has preferences

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) U(c_t, n_t), \qquad 0 < \beta < 1.$$

Technology

$$y_t = F(k_t, n_t, s_t), \qquad F_k > 0, \quad F_n > 0.$$

Output can be used for private consumption, government consumption, and investment.

Competitive production and factor prices

- Firms operate a neoclassical production technology and take factor prices as given.
- Under competitive pricing, before-tax factor prices equal marginal products:

$$r_t = F_k(k_t, n_t, s_t), \qquad w_t = F_n(k_t, n_t, s_t).$$

- Capital depreciates at rate δ. The after-tax gross return on capital is

$$R_t^k = 1 + (1 - \tau_t^k)(r_t - \delta).$$

- The government can tax labor income and capital income, but lump-sum taxation is ruled out.

Source of the distortion

The household's labor–leisure margin and intertemporal capital-allocation margin are affected by proportional income taxes.

Government instruments and budget constraint

- At history s^t, the government chooses

$$\pi_t = (\tau_t^n, \tau_t^k, R_t^b),$$

where τ_t^n and τ_t^k are labor- and capital-income tax rates, and R_t^b is the return on one-period government debt.

- Government debt is state contingent: the payoff can depend on the realization at s^{t+1}.

Government budget constraint

$$g_t + R_t^b b_t = \tau_t^n w_t n_t + \tau_t^k (r_t - \delta) k_t + b_{t+1}.$$

The initial stocks of capital and debt, k_0 and b_0, are given. Government consumption g_t is exogenous.

Household budget constraint

At each history s^t, the representative household chooses consumption, labor, next-period capital, and next-period government bonds:

$$x_t = (c_t, n_t, k_{t+1}, b_{t+1}).$$

Household budget constraint

$$\boxed{c_t + k_{t+1} + b_{t+1} \leq (1 - \tau_t^n) w_t n_t + R_t^b b_t + R_t^k k_t.}$$

The household takes prices and fiscal policy as given and maximizes expected lifetime utility. Consumption and labor are constrained to be nonnegative.

Household optimization: first-order conditions

Let $\lambda_t \geq 0$ denote the multiplier on the period-t household budget constraint. The FOCs are:

$$
\begin{aligned}
c_t: \quad & \beta^t \mu(s^t) U_{c,t} \leq \lambda_t, && \text{equality if } c_t > 0, \\
n_t: \quad & \beta^t \mu(s^t) U_{n,t} \leq -\lambda_t (1 - \tau_t^n) w_t, && \text{equality if } n_t > 0.
\end{aligned}
$$

For asset choices, the Euler conditions are

$$
\begin{aligned}
b_{t+1}: \quad & \left[\sum_{s^{t+1}} \lambda_{t+1} R_{t+1}^b - \lambda_t \right] b_{t+1} = 0, \\
k_{t+1}: \quad & \left[\sum_{s^{t+1}} \lambda_{t+1} R_{t+1}^k - \lambda_t \right] k_{t+1} = 0.
\end{aligned}
$$

- These conditions, together with the budget constraints and transversality conditions, characterize household optimization (under the conditions invoked in the notes).

Transversality conditions and implementability

For every infinite history s^∞, the household's transversality conditions are

$$\lim_{t\to\infty} \lambda_t b_{t+1} = 0, \qquad \lim_{t\to\infty} \lambda_t k_{t+1} = 0.$$

Multiply the household budget constraint by λ_t, sum over dates and histories, and use the asset Euler conditions and TVCs. The intertemporal household budget identity becomes

$$\sum_{t=0}^{\infty} \sum_{s^t} \lambda_t \left[c_t - (1-\tau_t^n) w_t n_t \right] = \lambda_0 (R_0^b b_0 + R_0^k k_0).$$

Using the consumption and labor FOCs (with interiority), obtain the implementability constraint:

Implementability constraint (IC)

$$\boxed{\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) \left(U_{c,t} c_t + U_{n,t} n_t \right) = U_{c,0} (R_0^b b_0 + R_0^k k_0).}$$

Aggregate feasibility

Adding the household and government budget constraints eliminates taxes and debt payments:

Aggregate resource constraint (RC)

$$c_t + g_t + k_{t+1} \leq F(k_t, n_t, s_t) + (1 - \delta)k_t.$$

- The resource constraint is a technological feasibility condition.
- The implementability constraint summarizes the restrictions imposed by household optimization and the available fiscal instruments.
- Together, they form the key constraints of the primal Ramsey allocation problem.

Planning Problem

Ramsey equilibrium

A Ramsey equilibrium consists of a policy π, an allocation rule $x(\cdot)$, and price rules $w(\cdot)$ and $r(\cdot)$ such that:

1. The government chooses π to maximize household welfare, taking the policy-dependent allocation and price rules as given.
2. For every candidate policy π', households solve their utility-maximization problem subject to their budget constraints.
3. For every π', factor prices satisfy the competitive marginal-product conditions.

Initial-period convention

The initial capital tax rate τ_0^k and initial government-debt return R_0^b are fixed. This prevents the government from exploiting the initial capital stock through an unrestricted surprise capital levy.

The Ramsey allocation problem

Proposition: primal characterization

The consumption, labor, and capital allocations in a Ramsey equilibrium solve

$$\max_{\{c_t, n_t, k_{t+1}\}} \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \mu(s^t) U(c_t, n_t)$$

subject to the implementability constraint

$$\sum_{t, s^t} \beta^t \mu(s^t)(U_{c,t} c_t + U_{n,t} n_t) = U_{c,0}(R_0^b b_0 + R_0^k k_0),$$

and the resource constraint

$$c_t + g_t + k_{t+1} \leq F(k_t, n_t, s_t) + (1 - \delta) k_t.$$

The tax rates and debt returns do not appear directly as choice variables in this primal formulation.

Recovering the bond allocation

Once the Ramsey allocation (c, n, k) is known, the household's intertemporal budget identity pins down the bond allocation:

$$b_{t'} = \frac{1}{U_{c,t'}} \sum_{j=t'+1}^{\infty} \sum_{s^j} \beta^{j-t'} \mu(s^j \mid s^{t'}) \big(U_{c,j} c_j + U_{n,j} n_j\big) - k_{t'}.$$

- The bond position is a residual object implied by the chosen allocation and the household's intertemporal budget constraint.
- It is not an independent degree of freedom once the Ramsey allocation and initial conditions are fixed.

Primal transformation

The government problem can be studied through (c, n, k) first; the remaining decentralized objects are reconstructed afterward.

Lagrangian representation

Let η be the multiplier on the implementability constraint and define

$$W(c_t, n_t, \eta) = U(c_t, n_t) + \eta\big[U_{c,t}c_t + U_{n,t}n_t\big].$$

The Ramsey problem can be written as maximizing

$$\sum_{t,s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) - \eta U_{c,0}(R_0^k k_0 + R_0^b b_0)$$

subject to the resource constraint. Its Lagrangian is

$$\begin{aligned}\mathcal{L} = &\sum_{t,s^t} \beta^t \mu(s^t) W(c_t, n_t, \eta) \\ &+ \sum_{t,s^t} \theta_t \big[F(k_t, n_t, s_t) + (1-\delta)k_t - c_t - g_t - k_{t+1}\big] \\ &- \eta U_{c,0}(R_0^k k_0 + R_0^b b_0),\end{aligned}$$

where $\theta_t \geq 0$ is the multiplier on the resource constraint.

Ramsey first-order conditions: $t \geq 1$

For periods $t \geq 1$, the FOCs are

$$\begin{aligned} c_t: &\quad W_{c,t} = \theta_t, \\ n_t: &\quad W_{n,t} = -\theta_t F_{n,t}, \\ k_{t+1}: &\quad \theta_t = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) \theta_{t+1}(1 - \delta + F_{k,t+1}). \end{aligned}$$

Eliminating θ_t gives the two central conditions:

Intratemporal condition

$$-\frac{W_{n,t}}{W_{c,t}} = F_{n,t}.$$

Intertemporal Euler equation

$$W_{c,t} = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) W_{c,t+1}(1 - \delta + F_{k,t+1}).$$

Why period 0 is different

The implementability constraint contains initial wealth, and the initial capital return depends on the initial allocation. Consequently, the period-0 FOCs contain additional terms.

$$-\frac{W_{n,0} - \eta\left[U_{cn,0}(R_0^k k_0 + R_0^b b_0) + U_{c,0}(1-\tau_0^k)F_{kn,0}\right]}{W_{c,0} - \eta U_{cc,0}(R_0^k k_0 + R_0^b b_0)} = F_{n,0},$$

$$W_{c,0} - \eta U_{cc,0}(R_0^k k_0 + R_0^b b_0) - \sum_{s_1} \beta\mu(s_1 \mid s_0) W_{c,1}(1 - \delta + F_{k,1}) = 0.$$

Markov structure

If s_t is Markov, then from period 1 onward the allocation rules are stationary functions

$$c(k, s; \eta), \quad n(k, s; \eta), \quad k_+(k, s; \eta), \quad b(k, s; \eta).$$

The period-0 rules differ because of the initial-wealth terms.

Recovering Ramsey policies

Given the Ramsey allocation, competitive pricing and household FOCs imply

Labor–consumption condition

$$-\frac{U_{n,t}}{U_{c,t}} = (1 - \tau_t^n) F_{n,t}.$$

The two household asset Euler equations are

$$U_{c,t} = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) U_{c,t+1} R_{t+1}^k,$$

$$U_{c,t} = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) U_{c,t+1} R_{t+1}^b.$$

- The labor tax is determined by the intratemporal condition.
- The capital tax and government-debt return are linked through two Euler equations, but are not separately pinned down state by state.

Policy indeterminacy: an equivalence class

Proposition 2.3 (in the notes)

If (R^b, τ^k) support a given allocation, then alternative policies $(\bar{R}^b, \bar{\tau}^k)$ also support it provided that, at each current history,

$$\sum_{s^{t+1}} \mu(s^{t+1} \mid s^t) U_{c,t+1} R^b_{t+1} = \sum_{s^{t+1}} \mu(s^{t+1} \mid s^t) U_{c,t+1} \bar{R}^b_{t+1},$$

$$\sum_{s^{t+1}} \mu(s^{t+1} \mid s^t) U_{c,t+1} \tau^k_{t+1} (F_{k,t+1} - \delta) = \sum_{s^{t+1}} \mu(s^{t+1} \mid s^t) U_{c,t+1} \bar{\tau}^k_{t+1} (F_{k,t+1} - \delta),$$

$$\tau^k_{t+1}(F_{k,t+1} - \delta) k_{t+1} - R^b_{t+1} b_{t+1} = \bar{\tau}^k_{t+1}(F_{k,t+1} - \delta) k_{t+1} - \bar{R}^b_{t+1} b_{t+1}.$$

If there are N possible next-period states, there are $N - 1$ degrees of indeterminacy at each node.

What is and is not identified?

Not uniquely determined

- Ex-post capital tax τ^k_{t+1} in each state.
- State-contingent government-debt return R^b_{t+1} in each state.

Pinned down by the allocation

- The allocation and bond positions.
- The relevant conditional expectations of asset returns and tax liabilities.
- The ex-ante capital tax rate.

Two polar restrictions that select a particular policy representation are:

1. Restrict the capital tax to be non-contingent on the next state: $\tau^k(s^t, s_{t+1}) = \bar{\tau}^k(s^t)$.
2. Restrict the return on government debt to be non-contingent on the next state: $R^b(s^t, s_{t+1}) = \bar{R}^b(s^t)$.

Ex-ante capital taxation and state prices

Define the one-period state price (SDF)

$$q_{t+1} = \beta \mu(s^{t+1} \mid s^t) \frac{U_{c,t+1}}{U_{c,t}}.$$

The ex-ante capital-income tax rate is

Ex-ante capital tax

$$\tau_t^e = \frac{\sum_{s^{t+1}} q_{t+1} \tau_{t+1}^k (F_{k,t+1} - \delta)}{\sum_{s^{t+1}} q_{t+1} (F_{k,t+1} - \delta)}.$$

- It is a state-price-weighted average of ex-post capital tax liabilities.
- The equivalence result leaves ex-post taxes indeterminate but pins down this ex-ante object.
- It is therefore important not to interpret τ_t^e as a state-by-state tax rate.

Debt returns and taxes on private assets

A state-non-contingent debt return $1 + \bar{r}_t$ can replicate the relevant state-contingent return through a tax on debt interest:

$$\sum_{s^{t+1}} q_{t+1} R^b_{t+1} = \sum_{s^{t+1}} q_{t+1}(1 + \bar{r}_t), \qquad R^b_{t+1} = 1 + \bar{r}_t(1 - \tau^b_{t+1}).$$

The effective tax on private assets combines capital-income and debt-income taxes:

Tax on private assets

$$\tau^a_{t+1} = \frac{\tau^k_{t+1}(F_{k,t+1} - \delta)k_{t+1} + \tau^b_{t+1}\bar{r}_t b_{t+1}}{(F_{k,t+1} - \delta)k_{t+1} + \bar{r}_t b_{t+1}}.$$

From period 1 onward, the notes describe the certain policy rules as functions of $(k, s; \eta)$: $\tau^n(k, s; \eta)$, $\tau^e(k, s; \eta)$, and $\tau^a(k, s; \eta)$.

An Analytical Result

Additive separability and zero ex-ante capital tax

The analytical result in CCK is derived for

$$U(c,n) = \frac{c^{1-\sigma}}{1-\sigma} + v(n), \qquad \sigma > 0,$$

with $U_{cn} = 0$ and logarithmic utility as the limit when $\sigma = 1$.

Proposition 3.1

The ex-ante capital-income tax is zero from period 1 onward:

$$\boxed{\tau_t^e = 0, \qquad \forall t \geq 1.}$$

- This concerns the ex-ante tax, not necessarily each state-contingent ex-post rate τ_{t+1}^k.
- The initial period is excluded because the initial capital tax is fixed by convention.
- The isoelastic consumption component is important for the proof: it makes W_c proportional to U_c.

Proof: compare the two Euler equations

For $t \geq 1$, the Ramsey planner's capital Euler equation can be written as

$$1 = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) \frac{W_{c,t+1}}{W_{c,t}} (1 - \delta + F_{k,t+1}). \tag{A}$$

The household's capital Euler equation is

$$1 = \sum_{s^{t+1}} \beta \mu(s^{t+1} \mid s^t) \frac{U_{c,t+1}}{U_{c,t}} \left[1 + (1 - \tau^k_{t+1})(F_{k,t+1} - \delta)\right]. \tag{B}$$

Since

$$W_{c,t} = U_{c,t} + \eta \left(U_{c,t} + c_t U_{cc,t} + n_t U_{cn,t}\right),$$

isoelastic consumption utility and separability imply

$$c_t U_{cc,t} = -\sigma U_{c,t}, \qquad U_{cn,t} = 0, \qquad W_{c,t} = \left[1 + \eta(1 - \sigma)\right] U_{c,t}.$$

Hence the proportionality factor is constant across states and dates, so

$$\frac{W_{c,t+1}}{W_{c,t}} = \frac{U_{c,t+1}}{U_{c,t}}.$$

Additive separability alone would not deliver this proportionality for an arbitrary $u(c)$.

Proof: implication for the ex-ante tax

Substituting the equality of marginal-utility growth rates into (A) and subtracting the planner's Euler equation from the household's gives

$$\sum_{s^{t+1}} q_{t+1}\tau^k_{t+1}(F_{k,t+1} - \delta) = 0,$$

where

$$q_{t+1} = \beta\mu(s^{t+1} \mid s^t)\frac{U_{c,t+1}}{U_{c,t}}.$$

By the definition of the ex-ante capital tax,

$$\tau^e_t = \frac{\sum_{s^{t+1}} q_{t+1}\tau^k_{t+1}(F_{k,t+1} - \delta)}{\sum_{s^{t+1}} q_{t+1}(F_{k,t+1} - \delta)},$$

so

Conclusion

$$\boxed{\tau^e_t = 0, \qquad t \geq 1.}$$

Quantitative Analysis

Preferences, technology, and shocks

KPR preferences:

$$U(c_t, n_t) = \frac{\left[c_t^{\gamma}(1-n_t)^{1-\gamma}\right]^{1-\sigma}}{1-\sigma}.$$

The baseline sets $\sigma = 1$, giving $U = \gamma \ln c + (1-\gamma)\ln(1-n)$. A high-risk-aversion alternative sets $\sigma = 9$.

Technology with trend growth:

$$y_t = k_t^{\alpha}\left[n_t \exp(\rho t + z_{t+1})\right]^{1-\alpha}, \qquad z_{t+1} = \rho_z z_t + \varepsilon_{z,t}.$$

Government consumption:

$$g_t = G\exp(\rho t + \widetilde{g}_t), \qquad \widetilde{g}_{t+1} = \rho_g \widetilde{g}_t + \varepsilon_{g,t}.$$

Innovations: $\varepsilon_{z,t} \sim N(0, \sigma_z^2)$ and $\varepsilon_{g,t} \sim N(0, \sigma_g^2)$.

Calibration

The non-stochastic stationary economy is used to select parameter values and match the following U.S. targets.

Preferences and technology

Parameter	Value
γ	0.25
σ (baseline)	1
β	0.98
α	0.34
δ	0.08
ρ	0.016

Shock processes

Parameter	Value
G	0.07
ρ_g	0.89
σ_g	0.07
ρ_z	0.81
σ_z	0.04

Alternative preference specification

The high-risk-aversion experiment sets $\sigma = 9$ while retaining the remaining baseline parameters.

Steady-state targets and initial conditions

Calibration targets

Moment	Target
Capital–output ratio, K/Y	2.71
Fraction of available time worked	0.23
Government spending–output ratio, G/Y	0.18
Debt–output ratio, B/Y	0.51

Benchmark policies

Policy	Value
Capital tax, τ^k	27.1%
Labor tax, τ^n	23.7%

Initial conditions

$$k_0 = 1.05, \qquad R_0^b b_0 = 0.20.$$

The initial conditions are taken from the balanced-growth path of the deterministic economy under the benchmark policies.

Computational strategy: allocation rules

1. **Detrend.** Transform the growing economy into a stationary one. In the notes, this transformation affects the discount factor and depreciation rate.
2. **Fix the implementability multiplier.** Guess η. For $t \geq 1$, solve for stationary decision rules as functions of (k, s), where $s = (z, \widetilde{g})$.

Given k, s, and $k' = k_+(k, s; \eta)$:

- Use the resource constraint and the intratemporal condition to express $c(k, s; \eta)$ and $n(k, s; \eta)$.
- Use the Euler equation to determine $k_+(k, s; \eta)$.
- At $t = 0$, use the period-0 FOCs and the resource constraint to determine c_0, n_0 as functions of k_1, and then determine k_1.

Computational strategy: the bond rule

For $t \geq 1$, the bond allocation is also a stationary function of (k, s) for a given η. Writing $\bar{s} = (k, s)$, the recursive bond equation is

$$U_c(\bar{s})b(\bar{s}) = \sum_{s'} \beta\mu(s' \mid s)\Big[U_c(\bar{s}')c(\bar{s}') + U_n(\bar{s}')n(\bar{s}') + U_c(\bar{s}')b(\bar{s}') + U_c(\bar{s}')k'(\bar{s}') - U_c(\bar{s})k'(\bar{s})\Big].$$

- The stationary bond rule is a fixed point of the mapping implied by this equation.
- The paper restricts $k'(k, s)$ to a finite-parameter family and enforces the Euler equation at a finite set of points.
- The bond rule is approximated as continuous and piecewise linear in k for each fixed s, with the recursive equation imposed at a finite set of points.

Computational strategy: determine η and recover policy

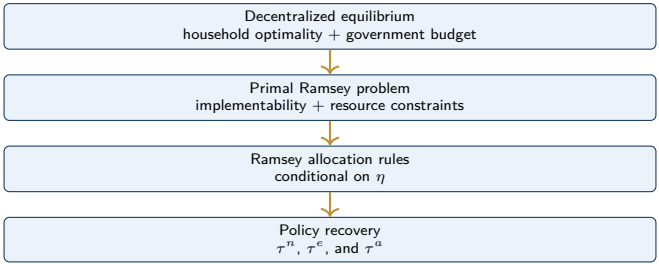
3. **Solve for the implementability multiplier.**
   - Substitute the period-0 rules into the household budget constraint.
   - Use the household intratemporal condition and set $\tau_0^k = 27.1\%$ to calculate the initial value $R_0^b b_0$.
   - Iterate on η until $R_0^b b_0 = 0.20$ is satisfied.
4. **Recover Ramsey policy rules.** Substitute the allocation rules into

$$-\frac{U_n}{U_c} = (1-\tau^n)F_n, \qquad \tau^e = \frac{\sum q\tau^k(F_k-\delta)}{\sum q(F_k-\delta)}, \qquad \tau^a = \frac{\tau^k(F_k-\delta)k + \tau^b\bar{r}b}{(F_k-\delta)k + \bar{r}b}.$$

Output of the numerical procedure

Stationary allocation rules and the associated labor-tax, ex-ante capital-tax, and private-asset-tax rules. The numerical outcomes themselves are presented separately.

Takeaway: the model-to-computation pipeline



The implementability constraint links household optimality to the primal planner problem; fiscal instruments are recovered only after the allocation is solved.

Main findings

Labor income tax

The labor tax fluctuates very little. Its persistence largely inherits the persistence of the exogenous shocks.

Ex-ante capital tax

It is identically zero under log utility and close to zero in the high-risk-aversion model.

Tax on private assets

Its mean is close to zero in the benchmark, but its standard deviation is large.

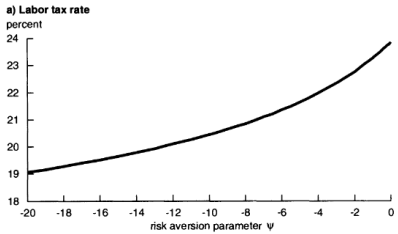
Interpretation

The first two instruments are smoothed across the business cycle; state-contingent taxes on private assets and debt returns absorb much of the fiscal impact of shocks.

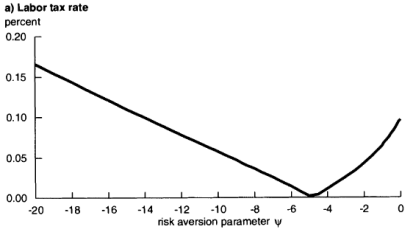
CCK (1994), Section VI.A. These are quantitative findings, not general theoretical restrictions on every decentralized tax instrument.

Risk-aversion sensitivity: labor income tax

Mean labor tax rate



Standard deviation

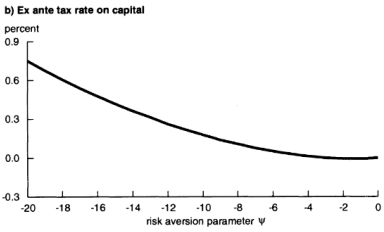


- As ψ becomes more negative (lower intertemporal substitutability), the mean labor tax declines.
- Its volatility is non-monotone and is close to zero around $\psi \simeq -4$; the correlation between the labor tax and shocks changes sign in this neighborhood.

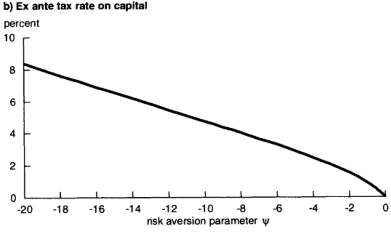
Figures 2(a) and 3(a): means and standard deviations as the risk-aversion parameter varies from 0 to -20.

Risk-aversion sensitivity: ex-ante capital tax

Mean ex-ante capital tax



Standard deviation



- The mean ex-ante tax remains below one percent even at $\psi = -20$ in this sensitivity exercise.
- Its standard deviation rises as ψ becomes more negative: a lower intertemporal elasticity makes variation in capital taxes less costly.

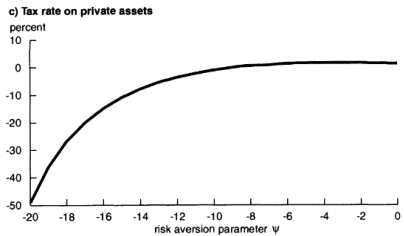
Robustness qualification

With i.i.d. shocks, high risk aversion, and baseline initial debt, the standard deviation can be large: about 25% at $\psi = -8$ and about 70% at $\psi = -20$ (the mean at $\psi = -20$ is 2.7%).

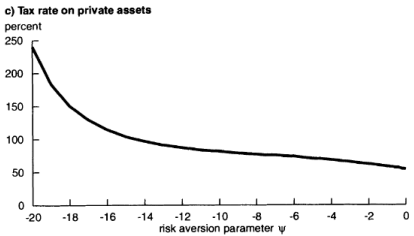
Figures 2(b) and 3(b); robustness experiment discussed by CCK (1994), Section VI.A.

Risk-aversion sensitivity: tax on private assets

Mean tax on private assets



Standard deviation

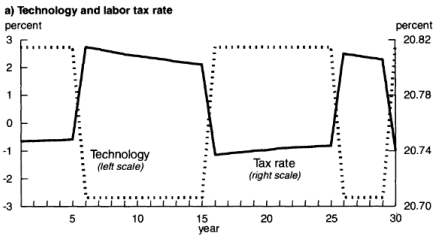


- The mean tax on private assets falls as ψ becomes more negative. CCK relate this to the increasingly negative steady-state debt position associated with a larger transition-period capital tax.
- The tax remains highly volatile relative to the labor and ex-ante capital tax rates.

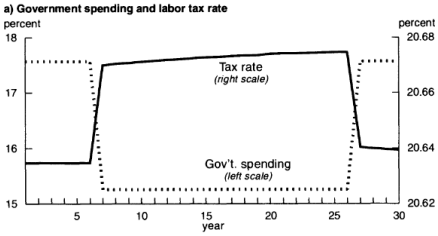
Figures 2(c) and 3(c). The tax on private assets combines the fiscal wedge on private capital and government debt; it is not a uniquely identified ex-post capital tax rate.

Simulated dynamics: labor income tax

Technology shocks only



Government-spending shocks only

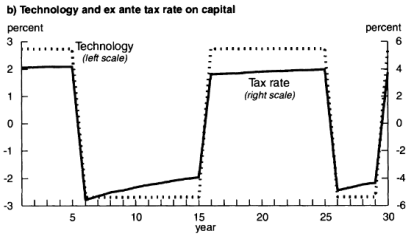


- The high-risk-aversion simulation shows discrete changes in the labor tax when the underlying shock changes, followed by relative stability between innovations.
- In this high-risk-aversion model, the labor tax rises when technology falls and when government spending falls. CCK report that these response directions are reversed in the baseline model.

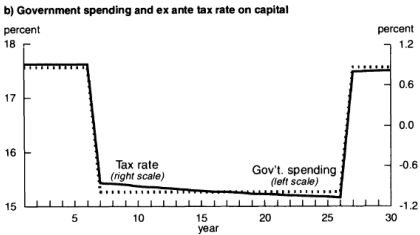
Figures 5(a) and 6(a): 30 simulated observations; high-risk-aversion model.

Simulated dynamics: ex-ante capital tax

Technology shocks only



Government-spending shocks only

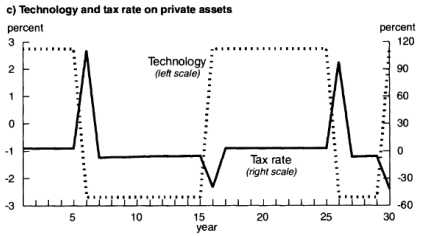


- The ex-ante capital tax responds around shock changes but is comparatively stable between innovations.
- This is consistent with the finding that the expected capital-tax wedge is small in the stationary equilibrium; it does not imply that every state-contingent ex-post tax is zero.

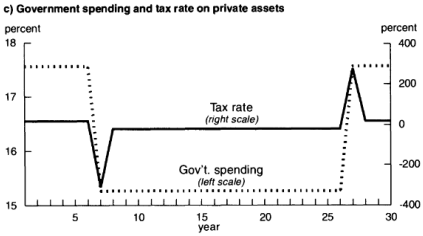
Figures 5(b) and 6(b): high-risk-aversion model.

Simulated dynamics: tax on private assets

Technology shocks only



Government-spending shocks only



- A negative technology innovation or a positive government-spending innovation generates a positive innovation in the tax on private assets.
- The response is large and concentrated around the shock; the labor tax is comparatively insulated.

Figures 5(c) and 6(c): high-risk-aversion model.

State-contingent private-asset taxation as a shock absorber

Government-spending experiment

When government spending rises from 15.2% to 17.5% of output, the tax rate on private assets rises from approximately 0 to 300%.

Baseline model

83%

Share of the innovation in the expected present value of spending financed in the current period by private-asset-tax revenue.

High-risk-aversion model

68%

Corresponding share financed by private-asset-tax revenue.

The model therefore uses state-contingent capital-tax and debt-return instruments to absorb fiscal shocks, rather than requiring large movements in the labor tax. CCK (1994), discussion of Figure 6(c), pp. 641–642.

Welfare gains: the transition is quantitatively important

Preference / benchmark	Economy	Ramsey	Zero cap. tax	High cap. tax
Log utility	Deterministic	1.00	0.20	0.00
	Stochastic: estimated policy	1.00	0.20	0.08
	Stochastic: variable policy	1.60	0.80	0.60
High risk aversion	Deterministic	1.30	-0.06	0.00
	Stochastic: estimated policy	1.20	-0.02	0.03
	Stochastic: variable policy	1.60	0.33	0.38
High initial debt	Deterministic	5.20	-0.70	0.00
	Stochastic: estimated policy	5.10	-0.60	0.05
	Stochastic: variable policy	6.10	0.30	1.00

How to read the table (CCK, Table 5)

All entries are welfare gains in consumption-equivalent percentage terms relative to the corresponding benchmark. The policy experiments impose constant tax rates, either with zero capital taxation or with a high capital tax.

Where do the welfare gains come from?

1. **Initial capital taxation is the main source.** In the log-utility benchmark, the constant-tax system with zero capital taxes delivers a 0.2% gain versus a 1.0% Ramsey gain. Thus about 80% of the Ramsey gain comes from the large transition-period capital tax.
2. **The initial labor-tax adjustment matters little.** Holding the labor tax constant from period 0 while retaining the Ramsey capital-tax path yields welfare gains indistinguishable from those under the full Ramsey plan.
3. **Smoothing stationary tax rates has modest value under estimated policy volatility.** CCK report small gains from eliminating labor- and ex-ante-capital-tax fluctuations; when benchmark policy volatility is multiplied by five, the gain from smoothing rises substantially (almost 40% of the Ramsey gain in the log-utility case).

Important nonstationarity

A constant zero capital tax in every period can lower welfare in the high-risk-aversion and high-debt cases: it misses the large initial tax that finances the transition.

CCK (1994), Section VII, discussion of Table 5.

Quantitative takeaway

- **Stationary allocation:** labor taxes are smooth and the ex-ante capital-tax wedge is close to zero.
- **Across states:** state-contingent capital taxes and debt returns absorb a large share of fiscal shocks; the tax on private assets is the flexible shock absorber.
- **Along the transition:** a large initial capital tax accounts for most of the welfare improvement from moving to the Ramsey plan.

A final qualification

The ex-post capital-tax path is not uniquely identified without specifying the accompanying government-debt return policy. The robust objects are the Ramsey allocation, the ex-ante tax wedge, and the combined private-asset fiscal position.

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Questions are welcome.